

EDGE-DC MACHINE — COMPLETE FACTOR MAP & VALUE STACK

Every lever across BTC · AI/HPC · Edge · ESG · RWA · Tax · Financing · Compliance · Risk — v1.0

This is the whole board. The power/site buy-box is one column of it (\$2). Honesty convention: each factor is tagged [REAL] (bankable today), [GATED] (real but conditional on a specific unlock), or [THIN] (often claimed, weak here — do not build the pitch on it).

1. THE LOADS — how a megawatt makes money

One contracted megawatt can serve multiple loads; the PPA lets you arbitrage between them rather than bet on one.

- [REAL] Bitcoin mining — the floor. Monetizes every idle/interruptible MW instantly, no tenant required. Low \$/MW, hashprice-volatile, ~70-80% of opex is power -> only works on cheap power ($\leq \sim 5.5c$). Its superpower: it's curtailable, which unlocks the grid revenue in §3.
 - [GATED] AI/HPC colocation — the premium. 5-10x the revenue/MW of BTC, but tenant-gated: nothing until an anchor lease is signed, and it needs \$3-6M/MW of conversion capex. The value multiplier, not the base case.
 - [GATED] Edge compute — latency-sensitive workloads sited near population/fiber. Smaller footprints, higher \$/MW, but needs the right location (metro-adjacent + fiber) and demand. A subset of the AI story, not separate money unless the site fits.
 - [REAL] Load arbitrage — convert MW from BTC->AI as tenants sign; run BTC on the balance. The flexibility itself is the asset. Never leave a contracted MW dark.
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2. THE POWER — the input and the moat (the buy-box column)

Power, not capital, is the binding constraint in Georgia.

- [REAL] Acquired legacy below-market PPA (Barak-class $\leq 4c$ vs 7.53c GA industrial avg) — the primary moat.

- [REAL] Behind-the-meter natural-gas generation — GA gas is deregulated (AGL territory, buy the molecule from Gas South et al.); self-generate below grid delivered rate. The real cost lever in a state where you can't shop electricity.
 - [GATED] Behind-the-meter solar + BESS — shaves demand/peak charges; ITC only if solar has a locked begin-construction date (window closed for un-started new solar; standalone-storage ITC still open).
 - [REAL] Interconnect / deliverable MW — energized interconnect beats a cheaper site stuck in the queue. Timing is the constraint, not rate.
 - [REAL] Curtailment optionality — the ability to shed load on demand -> feeds §3.
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3. ESG / ENVIRONMENTAL LAYER — mostly a CAPITAL-ACCESS and TENANT lever, not a revenue line

The honest value of ESG here is that it opens ESG-mandated capital and wins renewable-mandated tenants, plus one real tax lever and one real grid-revenue line. It is NOT a big carbon-credit stream.

- [REAL] Green-compute premium — renewable-powered, water-efficient, low-PUE capacity commands tenant preference and premium from hyperscalers/enterprises under renewable mandates. The biggest real ESG lever, and it's a revenue/tenant lever, not a credit.
- [REAL] Demand-response / grid-services revenue — curtailable BTC load sells interruptibility/ ancillary services back to the grid. Legitimate income AND a genuine ESG story (you stabilize the grid, you're not just a drain). Underused; quantify it.
- [REAL] Water-efficient cooling as risk mitigation — GA has drought exposure and active community/ political pushback on data-center water use. Air/immersion cooling cuts water -> lowers permitting/ community risk AND is a real ESG differentiator. Defensive value.
- [REAL] OZ as impact narrative — investing in a designated Opportunity Zone IS an impact/ESG story ESG-mandated LPs can underwrite. The tax benefit (§5) and the impact framing are the same fact.
- [GATED] BESS ITC / tax-equity — standalone-storage ITC is alive; real money, but a tax structure (§5), not an ESG revenue line.
- [THIN] RECs / SRECs — Georgia has NO RPS compliance market, so these are voluntary-market only: low, volatile value. Don't model them as material.
- [THIN] Carbon credits (Verra/Puro) — a compute pod does not inherently generate sequestration or avoidance credits. Only real with an actual, verifiable methodology. Do not pitch as a stream.
- [THIN] 45Q — carbon-capture credit; requires actual CCS on site. Not applicable without it.

Net ESG read: frame ESG as how you unlock cheaper/broader capital and premium tenants and de-risk permitting — not as a carbon-credit cash register. That framing survives diligence; the other doesn't.

4. RWA / CAPITAL STRUCTURE — fund scale without selling equity or returning to a bank

The flywheel: seasoned revenue -> tokenized cash-flow instrument -> proceeds fund the next pod -> new pod generates revenue -> repeat. (Re-securitization / capital recirculation.)

- [GATED] Tokenize the contracted power / offtake / seasoned cash flow — NOT the volatile BTC yield. A 7-yr PPA-backed cash-flow stream is what an RWA instrument wraps. Gate: the receivable must be pledgeable.
 - [REAL] ERC-3643 permissioned security token — your existing issuance/whitelist/ONCHAINID stack.
 - [REAL] CMBS-style tranche waterfall — senior/mezz/pref over the cash-flow stream; already in Id-capital-contracts.
 - [REAL] REIT-style distribution — pro-rata to KYC'd holders; your reit-dividend-engine.
 - [REAL] Oracle attestation — meter readings on-chain prove the revenue backing the token.
 - [REAL] AML/KYC/Travel-Rule gate + BitGo custody boundary — mandatory before value moves; non-custodial preserved.
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5. TAX & STRUCTURE — the shelters and the capstone

- [REAL] Opportunity Zone — federal cap-gains -> \$0 on appreciation over a 10-yr hold. QROF (rural) variant gives a 30% basis step-up vs the standard 10% if the site qualifies rural.
- [REAL] GA data-center sales/use tax exemption (2018, still in force) — on qualifying equipment.
- [GATED] ITC / tax-equity partnership flips — BESS standalone-storage ITC (solar only with a locked begin-construction date). Requires licensed-CPA sign-off.
- [REAL] Depreciation / bonus depreciation — on the equipment basis; ordinary but material.
- [REAL] Entity architecture — Unykorn = tech/compliance rails (no lending, no custody); LD Capital = licensed lender; per-site SPVs; Wyoming domicile. The wall must be real, not cosmetic (§7).

6. FINANCING / LEVERAGE STACK — the capital sequence

- [REAL] Bridge / hard-money — fast money to take down a PPA now, refi out via RWA. Speed is the edge.
- [REAL] SBA via Loan Depot — 504 for owner-occupied; slow (30-60 days) -> refi/takeout, not the first close.
- [GATED] RWA raise — the scale capital; needs seasoned, attestable cash flow first.
- [GATED] Tax equity — funds the BESS/solar against the ITC.
- [GATED] Vendor finance — potentially from the builder (R&K) on the pods.
- [REAL] QOF/QROF — the OZ fund is the equity vehicle tying the hold to the tax-free exit.

7. COMPLIANCE / REGULATORY — the gates that must clear

- [GATED] Securities characterization — "technology fees, not securities compensation": the issuance-bps fee is transaction-based comp and threatens the not-a-broker-dealer position. Counsel must bless the fee stack before you publish or take a fee. (Open item.)
- [REAL] Entity wall — Unykorn-rails vs LD-Capital-lending must exist on paper, or the "we don't do lending" claim contradicts the M Helen construction deal.
- [REAL] AML/KYC/Travel Rule — before any value movement, no exceptions.
- [REAL] Accredited-only / Reg D 506(c) — for the token/RWA offering.
- [GATED] Energy & environmental — interconnection agreements, GA PSC large-load rules, air permits for gas gen, water, noise. The data-center backlash bills (HB1063/SB410) died in 2026 but the political risk is live and could return.

8. RISK REGISTER — what actually kills it

- [HIGH] AI anchor-tenant risk — the entire AI upside is tenant-gated; no lease, no conversion.
- [HIGH] Hashprice volatility — the BTC floor moves with BTC price + network difficulty.
- [HIGH] PPA assignability / pledgeability — no assignment kills the deal; no pledge kills the RWA takeout.
- [MED] Interconnection timing — queue delay strands the whole build.
- [MED] Power/gas price volatility — erodes the spread the model rests on.
- [MED] Securities/BD registration — the issuance-bps exposure; fixable with counsel now, expensive later.

- [MED] Environmental permitting / community opposition — water + noise + GA data-center backlash.
 - [MED] Counterparty trust — the sourcing/real-estate partners; structure so trust isn't required.
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9. HOW IT INTERLOCKS (one paragraph)

Cheap contracted power (§2) is the moat. Bitcoin (§1) monetizes it from day one and its curtailability throws off grid-services + green-compute ESG value (§3) that opens ESG-mandated capital and premium AI tenants. Landing an AI anchor (§1) multiplies revenue/MW 5-10x. The seasoned cash flow gets tokenized (§4) to fund the next pod without selling equity — the recirculation flywheel. The whole thing sits in an Opportunity Zone (§5) so the built-up equity exits federally tax-free on a 10-yr hold, while the RWA layer provides liquidity along the way so the quick-liquidity and long-hold goals stop fighting. Compliance (§7) gates every value movement and keeps you in the un-regulated, un-capital-intensive rails seat. Everything is downstream of the power.